The change of affordability and availability in the housing market for young adults in the Netherlands from 2005 until 2025.

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Set-up your environment

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.1     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.3     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become error
```

```r
library(rmarkdown)
library(yaml)
library(readr)
library(dplyr)
library(ggplot2)
library(geodata)
```

```
## Loading required package: terra
## terra 1.9.27
##
## Attaching package: 'terra'
##
## The following object is masked from 'package:tidyr':
##
##     extract
```

```r
library(terra)
library(janitor)
```

```
##
## Attaching package: 'janitor'
```

```
##
## The following object is masked from 'package:terra':
##
##     crosstab
##
## The following objects are masked from 'package:stats':
##
##     chisq.test, fisher.test
```

```
library(scales)
```

```
##
## Attaching package: 'scales'
##
## The following object is masked from 'package:terra':
##
##     rescale
##
## The following object is masked from 'package:purrr':
##
##     discard
##
## The following object is masked from 'package:readr':
##
##     col_factor
```

```
library(patchwork)
```

```
##
## Attaching package: 'patchwork'
##
## The following object is masked from 'package:terra':
##
##     area
```

```
library(tinytex)
```

Title Page

Include your names

Include the tutorial group number

Include your tutorial lecturer's name

Part 1 - Identify a Social Problem

The Dutch housing market has changed significantly over the past two decades. For young adults, finding suitable housing has become increasingly difficult due to rising house prices, higher rents, and a continuing housing shortage (Nederlands Jeugdinstituut, n.d.). As a result, many young people struggle to enter the housing market.

This research examines how housing affordability and availability for young adults in the Netherlands changed between 2005 and 2025. It focuses on trends in house prices, rental costs, incomes and housing shortages to understand the main developments during this period.

1.1 Describe the Social Problem

The Dutch housing market has become increasingly inaccessible for young adults, making the move toward independent living far more difficult. This is socially relevant because stable and affordable housing is a foundation for financial independence, mental well-being, and equal life opportunities. When young people cannot secure a home, they delay major life steps such as studying, working full-time, or forming households—effects that ripple through society. The OECD notes that the Netherlands is among the countries where young adults face the greatest barriers to affordable housing due to rising prices and persistent shortages (OECD, 2023).

A combination of sharply increasing house prices, higher rents, and limited supply has created a structural imbalance between demand and availability. These pressures fall disproportionately on first-time buyers and young renters, widening generational inequality: older households benefit from accumulated housing wealth, while younger cohorts struggle to enter the market at all.

Include the following:

- Why is this relevant?
- …

Part 2 - Data Sourcing

2.1 Load in the data

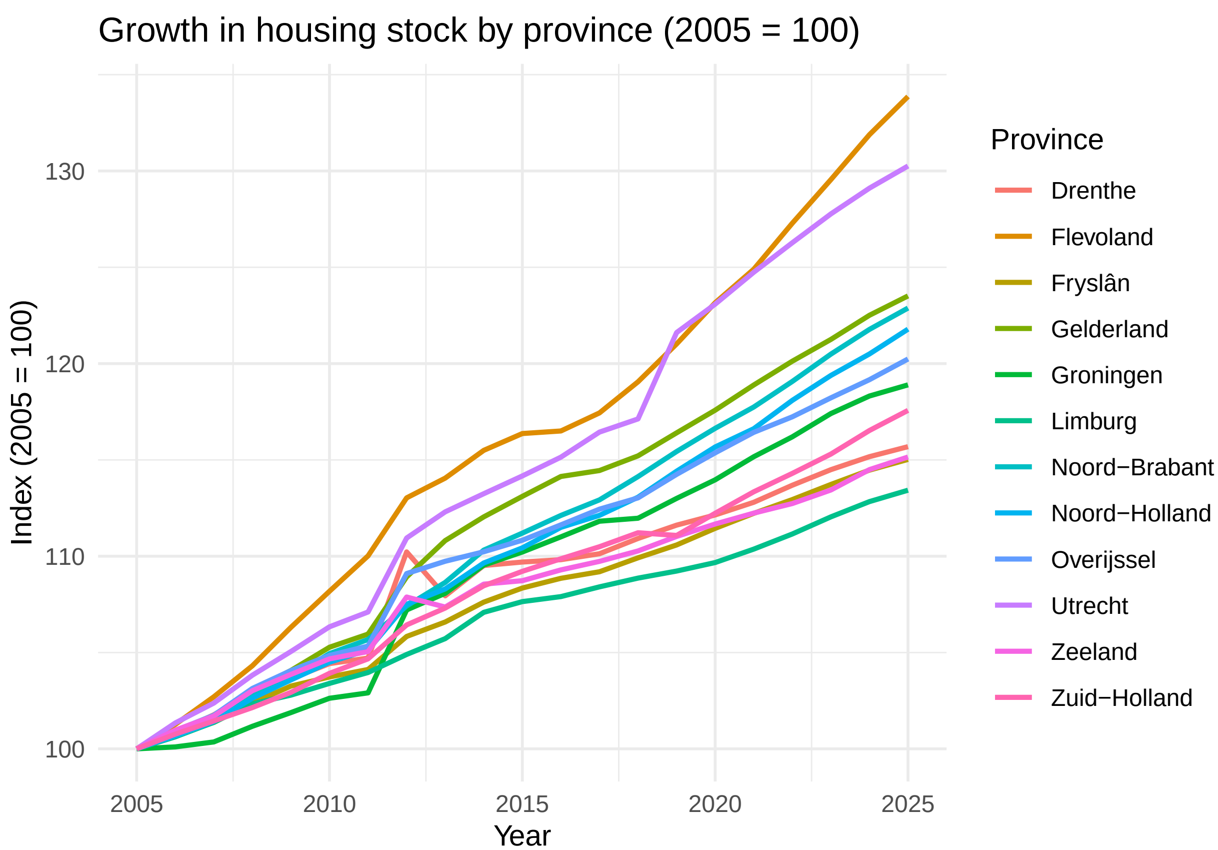
2.2 Provide a short summary of the dataset(s)

trendgrafieken per provincie Huizenprijzen/voorraad

```
df <- read_csv("data 2.0/Huizenprijzen_Huizenvoorraad_nieuw.csv") |>
  clean_names() |>
  mutate(provincie = str_remove(provincies, " \\(PV\\)"))
```

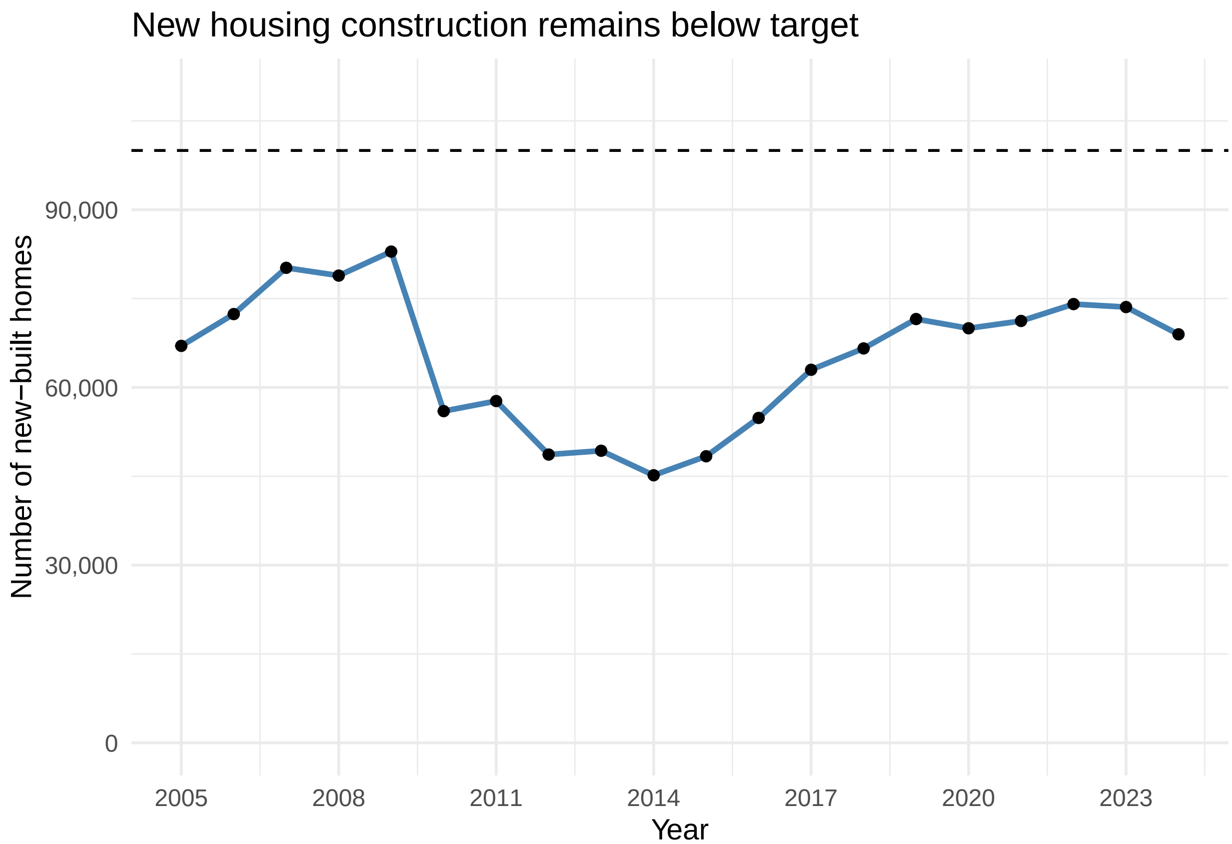
```
## Rows: 252 Columns: 4
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (1): Provincies
## dbl (3): Jaartal, Woningvoorraad_1_januari, Nieuwbouw
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
df_voorraad <- df |>
  group_by(provincie) |>
  arrange(jaartal) |>
  mutate(index = woningvoorraad_1_januari / first(woningvoorraad_1_januari) * 100) |>
  ungroup()

ggplot(df_voorraad, aes(jaartal, index, color = provincie)) +
  geom_line(linewidth = 0.9) +
  scale_x_continuous(breaks = seq(2005, 2025, 5)) +
  labs(title = "Growth in housing stock by province (2005 = 100)",
       x = "Year", y = "Index (2005 = 100)", color = "Province") +
  theme_minimal()
```

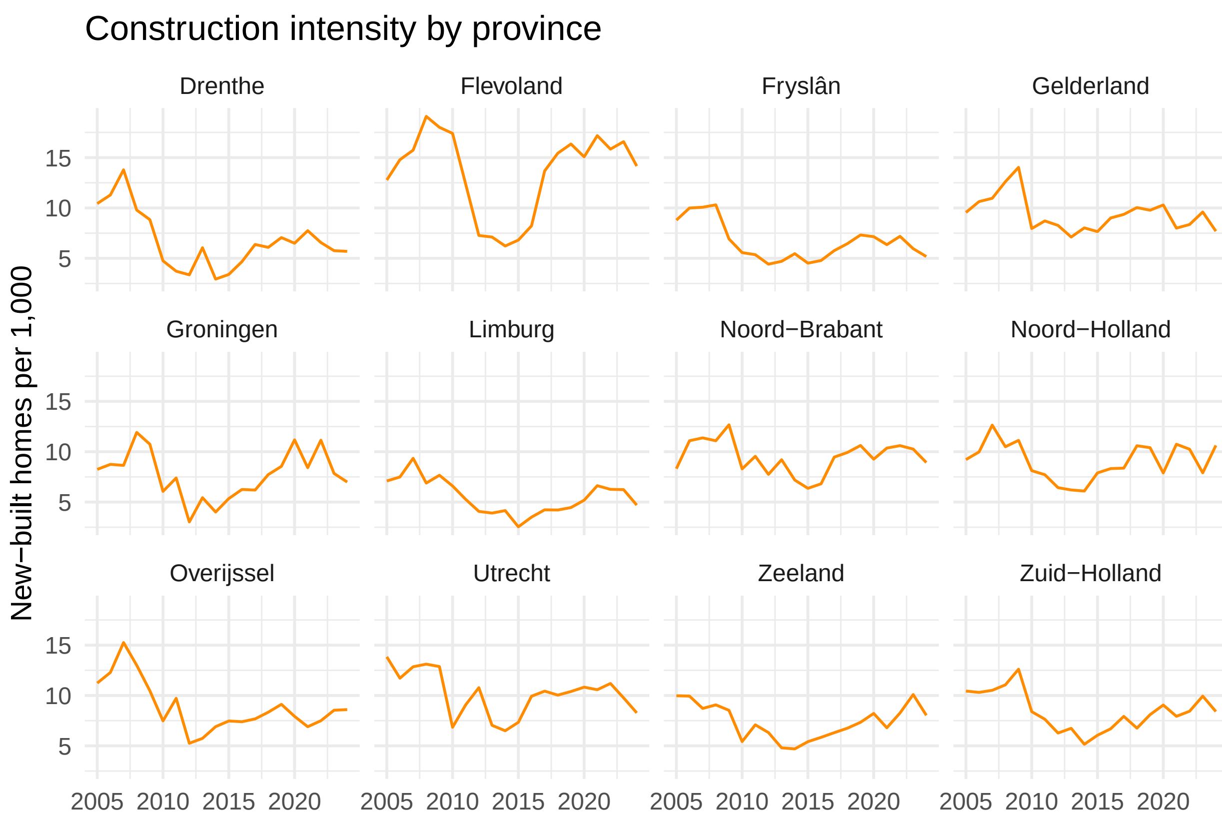



```r
df_nat <- df |>
  filter(jaartal < 2025) |>
  group_by(jaartal) |>
  summarise(nieuwbouw = sum(nieuwbouw, na.rm = TRUE))

ggplot(df_nat, aes(jaartal, nieuwbouw)) +
  geom_hline(yintercept = 100000, linetype = "dashed") +   # overheidsdoel
  geom_line(linewidth = 1, color = "steelblue") +
  geom_point() +
  scale_x_continuous(breaks = seq(2005, 2024, 3)) +
  scale_y_continuous(limits = c(0, 110000), labels = comma) +
  labs(title = "New housing construction remains below target",
```

```r
       x = "Year", y = "Number of new-built homes") +
  theme_minimal()
```




```r
df_intensiteit <- df |>
  filter(jaartal < 2025) |>
  mutate(per_1000 = nieuwbouw / woningvoorraad_1_januari * 1000)

ggplot(df_intensiteit, aes(jaartal, per_1000)) +
  geom_line(color = "darkorange") +
  facet_wrap(~ provincie) +
  scale_x_continuous(breaks = seq(2005, 2024, 5)) +
  labs(title = "Construction intensity by province",
       x = "", y = "New-built homes per 1,000") +
  theme_minimal()
```



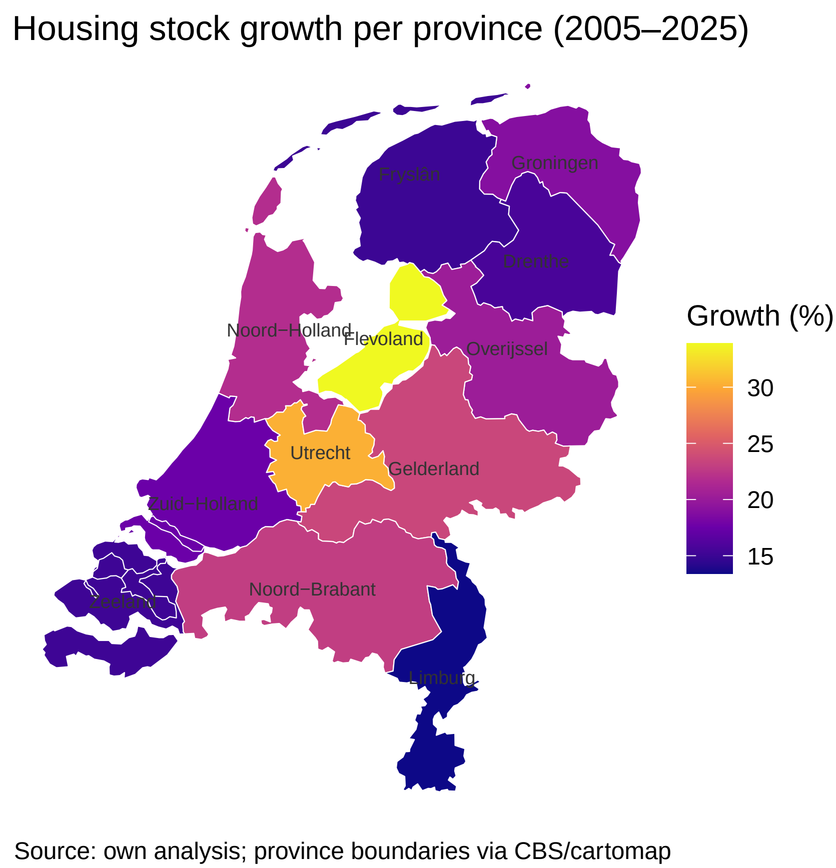
#Explanation I put the housing data into three charts, because housing stock and new construction are actually two different things. The stock is the total number of homes that exist, and new construction is what gets added each year. Putting those two in one chart with the same index would be a bit misleading, since the stock rises slowly while new construction jumps up and down a lot from year to year. In the first chart I set the stock per province to 2005 = 100, which makes the provinces comparable even though some are much bigger than others. In the second chart I added up new construction nationally per year and put a line at 100,000, the number the government wants to build each year. One caveat: my figures only cover actual new construction, while that target also counts things like offices turned into homes, so the line is an indicative lower bound and not an exact match. The third chart shows building intensity (new construction per 1,000 existing homes), which corrects for how big a province is. I left 2025 out for new construction, because that value is missing from the dataset, so I don't draw conclusions about that year.

#Conclusion Over time the housing stock grew steadily in every province, but new construction tells the more interesting story. After a peak around 2009 it dropped to a low of about 45,000 homes in 2014, and only climbed back to around 74,000 by 2022, never reaching the target of 100,000. The timing of that dip lines up with the 2008 credit crisis, but this is an association and not proof: the decline only set in with a delay (building has a long pipeline), and the later recovery was also shaped by other things like low interest rates and rising demand after 2020, so the crisis alone can't explain the whole pattern. It's also important to say that my data only measures supply, not the shortage itself. The link to demand comes from outside sources: the number of households has been growing faster than the housing stock (Rijksoverheid, n.d.), which left a statistical shortage of roughly 396,000 homes in 2025 (Volkshuisvesting Nederland, 2025). I can't show that shortage from my own data alone, but putting the two together suggests that for young adults the problem isn't only about prices, but also about availability: too few homes were added to keep up with the growing demand.

```
library(jsonlite)
```

```
##
## Attaching package: 'jsonlite'

## The following object is masked from 'package:purrr':
##
##     flatten
```

```r
groei <- df |>
  group_by(provincie) |>
  arrange(jaartal) |>
  summarise(groei_pct = (last(woningvoorraad_1_januari) / first(woningvoorraad_1_januari) - 1) * 100)

geo <- fromJSON("https://cartomap.github.io/nl/wgs84/provincie_2023.geojson",
                simplifyVector = FALSE)

rows <- list(); g <- 0
for (f in geo$features) {
  naam   <- f$properties$statnaam
  coords <- f$geometry$coordinates
  polys  <- if (f$geometry$type == "Polygon") list(coords) else coords  # alles -> lijst van polygonen
  for (poly in polys) for (ring in poly) {
    g <- g + 1
    m <- matrix(unlist(ring), ncol = 2, byrow = TRUE)   # ring -> matrix van [lon, lat]
    rows[[length(rows) + 1]] <- data.frame(provincie = naam, long = m[,1], lat = m[,2], group = g)
  }
}
prov_df <- do.call(rbind, rows)

# grenzen koppelen aan de groeicijfers
plotdata <- prov_df |> left_join(groei, by = "provincie")

labels <- plotdata |>
  group_by(provincie) |>
  summarise(long = mean(long), lat = mean(lat), .groups = "drop")

# choropleth tekenen
ggplot(plotdata, aes(long, lat, group = group, fill = groei_pct)) +
  geom_polygon(color = "white", linewidth = 0.2) +
  geom_text(data = labels, aes(long, lat, label = provincie),
            inherit.aes = FALSE, size = 2.3, color = "grey20") +
  coord_quickmap() +                                  # juiste verhouding zonder sf
  scale_fill_viridis_c(option = "C") +
  labs(title = "Housing stock growth per province (2005-2025)",
       caption = "Source: own analysis; province boundaries via CBS/cartomap",
       fill = "Growth (%)") +
  theme_void()
```



#Explanation To show where in the country the growth happened, I mapped the total growth of the housing stock per province between 2005 and 2025. For each province I worked out the percentage change in the stock and used that to colour the map. The province borders come from a public CBS/cartomap file that I read in as a GeoJSON, so I didn't need any extra mapping software. One small caveat: the province labels sit at the average of each province's border points, which is only a rough centre, so for oddly shaped provinces the label can land slightly off. The colours and the actual figures aren't affected by that. #Conclusion The map shows a clear centre–periphery pattern. Growth is highest in the central provinces Flevoland (+34%) and Utrecht (+30%) and lowest in the ones on the edges, with Limburg in the far southeast (+13%) and the coastal provinces Zeeland and Fryslân (around +15%) at the bottom. This matters for the research question, because the stock grew fastest exactly in the central regions where demand from students and young workers is highest, while the quieter provinces on the edges grew the least. So the new supply wasn't added where it would help the most, which helps explain why housing stayed hard to find for young adults in precisely the areas they most want to live.

```
#Here we modify the dataset Inflatie
inflatie <- read.csv("data/Inflatie.csv")
inflatie <- inflatie %>%
  rename(Jaartal = Perioden) %>%
  rename(Inflatie = Jaarmutatie.CPI....) %>%
  select(Jaartal,Inflatie)
write.csv(inflatie, "data 2.0/Inflatie_nieuw.csv", row.names = FALSE)
```

```
#Here we modify the dataset Welvaart_van_huishoudens__kerncijfers
welvaart <- read.csv("data/Welvaart_van_huishoudens__kerncijfers.csv")
welvaart <- welvaart %>%
  rename(leeftijd_kostwinner = Kenmerken.van.huishoudens) %>%
```

```
  rename(Jaartal = Perioden) %>%
  rename(Gemiddeld_gestandaardiseerd_inkomen_x1000_euro = Gemiddeld.gestandaardiseerd.inkomen..1.000.eu
head(welvaart)
```

```
##               leeftijd_kostwinner Jaartal
## 1 Hoofdkostwinner: 25 tot 45 jaar    2011
## 2 Hoofdkostwinner: 25 tot 45 jaar    2012
## 3 Hoofdkostwinner: 25 tot 45 jaar    2013
## 4 Hoofdkostwinner: 25 tot 45 jaar    2014
## 5 Hoofdkostwinner: 25 tot 45 jaar    2015
## 6 Hoofdkostwinner: 25 tot 45 jaar    2016
##   Gemiddeld_gestandaardiseerd_inkomen_x1000_euro
## 1                                           25.0
## 2                                           25.0
## 3                                           25.2
## 4                                           26.2
## 5                                           26.5
## 6                                           27.7
```

```
write.csv(welvaart, "data 2.0/Welvaart_van_huishoudens_kerncijfers_nieuw.csv", row.names = FALSE)
```

```
#Here we modify the dataset Bestaand_koopwoningen verkoopprijzen
Bestaande_koopwoningen_verkoopprijzen_woningtype <- read.csv("data/Bestaande_koopwoningen_verkoopprijzen
Bestaande_koopwoningen_verkoopprijzen_woningtype <- Bestaande_koopwoningen_verkoopprijzen_woningtype %>%
  rename( Jaartal = Perioden)
Bestaande_koopwoningen_verkoopprijzen_woningtype <- Bestaande_koopwoningen_verkoopprijzen_woningtype %>%
  rename(Aantal_verkochte_woningen = Verkochte.bestaande.koopwoningen.Verkochte.woningen..aantal.)
Bestaande_koopwoningen_verkoopprijzen_woningtype <- Bestaande_koopwoningen_verkoopprijzen_woningtype %>%
  rename( Gemiddelde_verkoopprijs_euro = Gemiddelde.verkoopprijs..euro.)
Bestaande_koopwoningen_verkoopprijzen_woningtype <- Bestaande_koopwoningen_verkoopprijzen_woningtype %>%
  select(Jaartal, Aantal_verkochte_woningen, Gemiddelde_verkoopprijs_euro)
  write.csv(Bestaande_koopwoningen_verkoopprijzen_woningtype, "data 2.0/Bestaande_koopwoningen_verkoopp
```

The next thing we are curious about is how the growth of the population of young adults in the Netherlands relate to the growth of housing. By comparing this two variables over time we hope to gain insight into why there is such a shortage of housing.

To get started we will change the dataset containing the information about population growth and rename the columns using the rename() function, so it is easier to comprehend. After that we only select the useful information via transmute(). And lastly we put the variable for time in the first columns to improve the look of the table.

```
#Here we modify the dataset bevolkingsgroei to only keep the relevant information and use the right yea
bevolkingsgroei <-read.csv("data/Bevolkingsgroei_opbouw_2005_2019.csv")
bevolkingsgroei <- bevolkingsgroei %>%
  rename( jong_volwassenen_20_tot_45_jaar = Bevolking.op.1.januari.Naar.leeftijd.20.tot.45.jaar..x.1.000
  transmute(jong_volwassenen_20_tot_45_jaar) %>%
  mutate(Jaar = 2005:2019) %>%
  relocate(Jaar) %>%
  rename(Jaartal = Jaar)

write.csv(bevolkingsgroei, "data 2.0/Bevolkingsgroei_nieuw.csv", row.names = FALSE)
```

The next part is dedicated to gather information about the number of new houses being built in the Netherlands. In the first part of code we again change collumn names and select the useful comments.

There is however a hurdle in this dataset that we have to overcome. That is that the number of new houses built is grouped per province, while we would like to look at the number of the entire Netherlands. To achieve this, we used the goup_by() and summarise() functions.

```
#Here we modify the dataset Huizenprijzen_Huizenvoorraad
huizenvoorraad <-read.csv("data/Huizenprijzen_Huizenvoorraad.csv")
huizenvoorraad <- huizenvoorraad %>%
  rename(Jaartal = Perioden) %>%
  rename(Provincies = Regio.s) %>%
  rename(Woningvoorraad_1_januari = Bouwen.en.wonen.Woningvoorraad.Voorraad.op.1.januari..aantal.)%>%
  rename(Nieuwbouw = Bouwen.en.wonen.Woningvoorraad.Nieuwbouw..aantal.) %>%
  select(Jaartal, Provincies, Woningvoorraad_1_januari, Nieuwbouw)

tabel_per_jaar <- huizenvoorraad %>%
  group_by(Jaartal) %>%
  summarise(
    Woningvoorraad_1_januari = sum(Woningvoorraad_1_januari, na.rm = TRUE),
    huizenvoorraad = sum(Nieuwbouw, na.rm = TRUE)
  )
  write.csv(tabel_per_jaar, "data 2.0/Huizenvoorraad_nieuw.csv", row.names = FALSE)
```

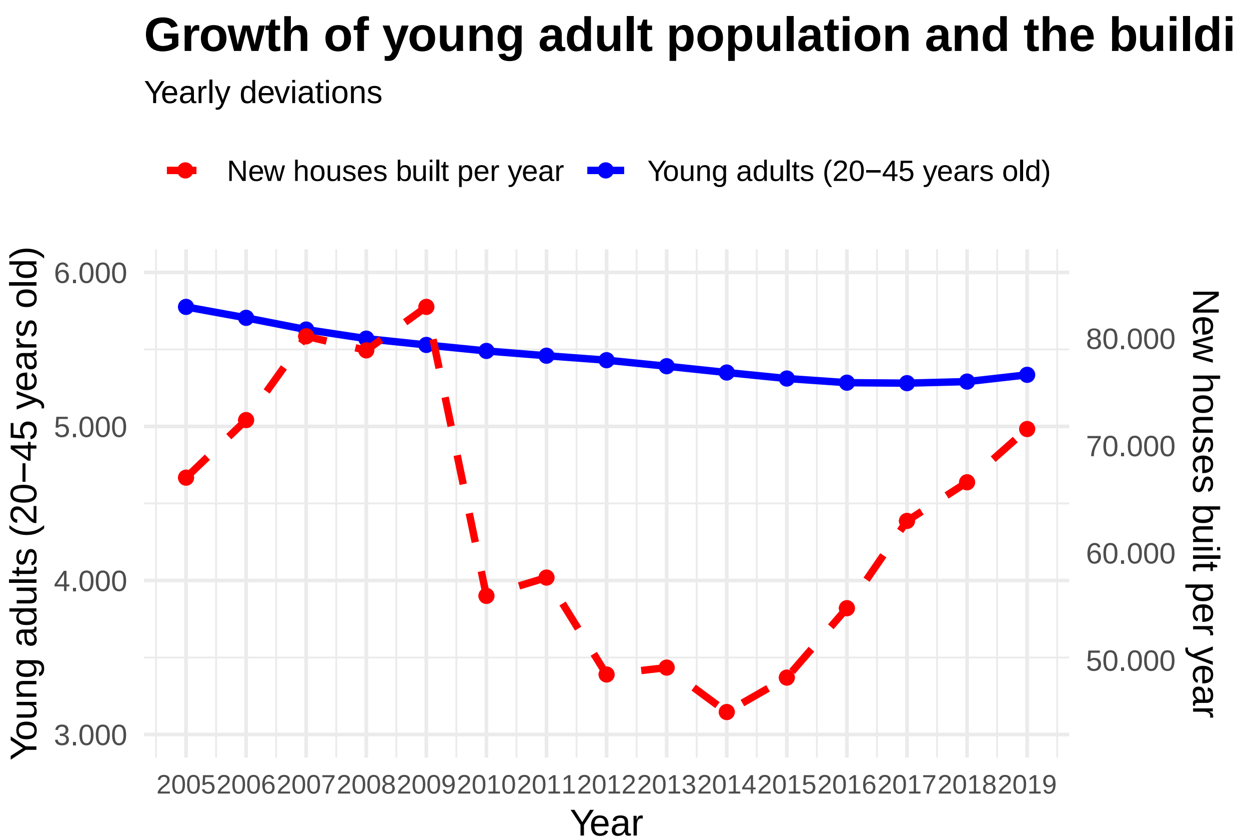
Now that we have two clear datasets we still need to merge them to be able to plot them into the same graph. We decided to join the two datasets on using years, since we want to look at the change over time and the trend that is going on. We used the left join to join the newly built housing dataset onto the population growth dataset. We did this because the first named spands across the least amount of years. Using the left join we ensure that we don't get Na's where we do not want them.

```
#Here we merge the datasets
bevolkingsgroei_nieuwbouw <- left_join(bevolkingsgroei, tabel_per_jaar, by = "Jaartal")
write.csv(bevolkingsgroei_nieuwbouw, "data 2.0/bevolkingsgroei_nieuwbouw")
```

```
#Here we create a graph to compare the amount of new houses built and the population growth of young ad

bevolkingsgroei_nieuwbouw <- read.csv("data 2.0/bevolkingsgroei_nieuwbouw")
# d
scale_factor <- max(bevolkingsgroei_nieuwbouw$jong_volwassenen_20_tot_45_jaar, na.rm = TRUE) /
                max(bevolkingsgroei_nieuwbouw$huizenvoorraad, na.rm = TRUE)

ggplot(bevolkingsgroei_nieuwbouw, aes(x = Jaartal)) +

  # Left axis(population growth)
  geom_line(
    aes(y = jong_volwassenen_20_tot_45_jaar,
        color = "Young adults (20-45 years old)"),
    linewidth = 1.3
  ) +
  geom_point(
    aes(y = jong_volwassenen_20_tot_45_jaar,
        color = "Young adults (20-45 years old)"),
    size = 2
  ) +
```

```r
# right axis(newly built homes)
geom_line(
  aes(y = huizenvoorraad * scale_factor,
      color = "New houses built per year"),
  linewidth = 1.3,
  linetype = "dashed"
) +
geom_point(
  aes(y = huizenvoorraad * scale_factor,
      color = "New houses built per year"),
  size = 2
) +
scale_y_continuous(
  #left axis
  name = "Young adults (20-45 years old)",
  limits = c(3000, 6000),
  labels = label_number(big.mark = ".", decimal.mark = ","),
  breaks = c(3000, 4000, 5000, 6000),

  #right axis
  sec.axis = sec_axis(
    ~ . / scale_factor,
    name = "New houses built per year",
    breaks = c(40000, 50000, 60000, 70000, 80000),
    labels = label_number(big.mark = ".", decimal.mark = ",")
  )
) +

scale_x_continuous(
  breaks = seq(min(bevolkingsgroei_nieuwbouw$Jaartal),
               max(bevolkingsgroei_nieuwbouw$Jaartal),
               by = 1)
) +

scale_color_manual(
  values = c(
    "Young adults (20-45 years old)" = "blue",
    "New houses built per year" = "red"
  )
) +

labs(
  title = "Growth of young adult population and the building of new houses",
  subtitle = "Yearly deviations",
  x = "Year",
  color = NULL
) +

theme_minimal(base_size = 14) +

theme(
  plot.title = element_text(face = "bold", size = 18),
  plot.subtitle = element_text(size = 12),
```

```
    axis.text.x = element_text(size = 10),
    legend.position = "top"
  )
```




```
inline_code = TRUE
```

```
#Here we modify the dataset Huizenprijzen_verschillende_woningtypes
huizenprijzen <- read_csv("data/Huizenprijzen_verschillende_woningtypes.csv")
```

```
## Rows: 147 Columns: 4
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (1): Type.woning
## dbl (3): Perioden, Prijsindex.bestaande.koopwoningen.Prijsindex.verkoopprijz...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
names(huizenprijzen)[names(huizenprijzen) == "Perioden"] <- "Jaren"
names(huizenprijzen)[names(huizenprijzen) == "Type.woning"] <- "Woningtype"
names(huizenprijzen)[names(huizenprijzen) == "Prijsindex.bestaande.koopwoningen.Prijsindex.verkoopprijz
names(huizenprijzen)[names(huizenprijzen) == "Prijsindex.bestaande.koopwoningen.Ontwikkeling..t.o.v..eer
write.csv(huizenprijzen, "data 2.0/nieuw_Huizenprijzen_per_woningtype.csv" , row.names = FALSE)
```

2.3 Describe the type of variables included

Think of things like:

- Do the variables contain health information or SES information?
-

  ### Have they been measured by interviewing individuals or is the data coming from administrative sources?

  #### Description and Data Collection

  The variables included in our project are **quantitative** in nature. They consist of continuous economic indicators—such as monetary values (income, house prices) and percentages (inflation)—as well as discrete metrics tracking physical counts (population size, housing stock). Categorical (nominal) variables such as `Provincies` and `Woningtype` are utilized to allow for spatial and structural segmentation.

Unlike survey data which relies on self-reported interviews, these datasets are sourced entirely from **administrative registries** compiled by Statistics Netherlands (CBS). Specifically, the socio-economic and income data is derived from the Dutch Tax Authority (Belastingdienst), while the housing metrics are registered through the Municipal Base Registry for Housing (BAG) and the Dutch Land Registry (Kadaster). This administrative nature ensures high data coverage and eliminates traditional interview biases such as recall or non-response bias.

Stated limitations of this data

While these administrative datasets are highly reliable, we face several limitations when merging and analyzing them to study young adults:

Inconsistent Demographic Definitions: The datasets categorize "young adults" differently. The income datasets slice data by the age of the primary wage earner (`leeftijd_kostwinner`), whereas the demographic dataset tracks individuals aged 20 to 45 (`jong_volwassenen_20_tot_45_jaar`). This structural mismatch limits our ability to perfectly isolate a single, uniform age bracket. **Temporal Truncation:** The demographic tracking for population build-up (`Bevolkingsgroei_opbouw.csv`) is currently limited to the 2005–2019 period within our data folder. This creates a data gap for the final years of our 2005–2025 analysis window. **Level of Aggregation:** Because the data is highly aggregated by the CBS to protect privacy, we can only join these files on the macro-level using the temporal dimension (`Jaartal`/`Jaren`) and the regional dimension (`Provincies`). We lack micro-data, meaning we cannot track individual household trajectories or specific cross-sections over time. —

For the sake of this example, I will continue with the assignment…

Part 3 - Quantifying

3.1 Data cleaning

Say we want to include only larger distances (above 2) in our dataset, we can filter for this.

```
#Start joining dataset inflatie_nieuw and Welvaart_van_huishoudens_kerncijfers_nieuw
welvaart <- read_csv("data 2.0/Welvaart_van_huishoudens_kerncijfers_nieuw.csv")
```

```
## Rows: 14 Columns: 3
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (1): leeftijd_kostwinner
## dbl (2): Jaartal, Gemiddeld_gestandaardiseerd_inkomen_x1000_euro
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
inflatie <- read_csv("data 2.0/Inflatie_nieuw.csv")
```

```
## Rows: 21 Columns: 2
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## dbl (2): Jaartal, Inflatie
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
result <- left_join(welvaart, inflatie, by = "Jaartal")
write.csv2(result,"data 2.0/Samenvoeging1.csv" , row.names = FALSE)
```

```
#Start joining dataset Samenvoeging1 and bruto_en_besteedbaar_inkomen_leeftijdsgroep_nieuw.csv
samenvoeging <- read_csv2("data 2.0/Samenvoeging1.csv")
```

```
## i Using "','" as decimal and "'.'" as grouping mark. Use `read_delim()` for more control.
```

```
## Rows: 14 Columns: 4
## -- Column specification --------------------------------------------------------
## Delimiter: ";"
## chr (1): leeftijd_kostwinner
## dbl (3): Jaartal, Gemiddeld_gestandaardiseerd_inkomen_x1000_euro, Inflatie
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
brutoinkomen <- read_csv("data 2.0/bruto_en_besteedbaar_inkomen_leeftijdsgroep_nieuw.csv")
```

```
## Rows: 14 Columns: 9
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (1): Kenmerken_van_huishoudens
## dbl (8): Jaartal, Particuliere_huishoudens_x1000, Totaal_bruto_inkomen_mln.e...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
samenvoeging2 <- left_join(brutoinkomen, samenvoeging, by = "Jaartal")
write.csv(samenvoeging2, "data 2.0/Samenvoeging2.csv" , row.names = FALSE)
head(samenvoeging2)
```

```
## # A tibble: 6 x 12
##   Jaartal Kenmerken_van_huishoud~1 Particuliere_huishou~2 Totaal_bruto_inkomen~3
##     <dbl> <chr>                                     <dbl>                  <dbl>
## 1    2011 Hoofdkostwinner: 25 tot~                  2463.                 160185
## 2    2012 Hoofdkostwinner: 25 tot~                  2445.                 159823
## 3    2013 Hoofdkostwinner: 25 tot~                  2422.                 159240
## 4    2014 Hoofdkostwinner: 25 tot~                  2394.                 161606
## 5    2015 Hoofdkostwinner: 25 tot~                  2367.                 157879
## 6    2016 Hoofdkostwinner: 25 tot~                  2348.                 160886
## # i abbreviated names: 1: Kenmerken_van_huishoudens,
## #   2: Particuliere_huishoudens_x1000, 3: Totaal_bruto_inkomen_mln.euro
## # i 8 more variables: Gemiddeld_bruto_inkomen_1000.euro <dbl>,
## #   Mediaan_bruto_inkomen_1000.euro <dbl>,
## #   Totaal_besteedbaar_inkomen_mln.euro <dbl>,
## #   Gemiddeld_besteedbaar_inkomen_1000.euro <dbl>,
## #   Mediaan_besteedbaar_inkomen_1000.euro <dbl>, leeftijd_kostwinner <chr>, ...
```

```r
#Deleting a column that was mentioned twice
samenvoeging2 <- read.csv("data 2.0/Samenvoeging2.csv")
samenvoeging2 <- samenvoeging2 %>%
  select (-leeftijd_kostwinner)
write.csv(samenvoeging2, "data 2.0/Final_dataset.csv", row.names = FALSE)
```

Please use a separate 'R block' of code for each type of cleaning. So, e.g. one for missing values, a new one for removing unnecessary variables etc.

3.2 Generate necessary variables

The first, necessary variable added is the maximal nominal mortgage loan that young adults can loan at the bank. The maximal nominal mortgage loan is based on the LTI and the median gross income. The LTI (Loan-to-Income) is the ratio between the gross income and the mortgage loan. This ratio decides the maximum amount that can be borrowed, which usually is around 4.5 times the gross income in the Netherlands (Homefinance Hypotheken, n.d.). The choice to use the median gross income over the average gross income is supported by the fact that the mean usually is distorted by high extremes, whilst the median gives a representative reflection of the "typical" young adult.

The Netherlands has one of the highest mortgage debts of Europe, which states that the Dutch housing market is really dependent on credits (DNB, n.d.). Because the Dutch housing market for buyers is fundamentally credit-driven and reliant on the LTI, variables solely addressing income are not enough to model the maximum purchasing capacity of young adults. However, this nominal borrowing ceiling is the variable that can demonstrate the maximum buying capacity of young adults. Moreover, this variable can be rightfully compared to the housing prices. All in all, the maximal nominal mortgage loan more or less serves as the baseline for the study regarding housing affordability.

```r
af_finaldataset <- read_csv("data 2.0/Final_dataset.csv")
```

```
## Rows: 14 Columns: 11
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr  (1): Kenmerken_van_huishoudens
## dbl (10): Jaartal, Particuliere_huishoudens_x1000, Totaal_bruto_inkomen_mln....
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
af_newvariable <- af_finaldataset %>%
  mutate(Max_nominal_mortgage_loan_1000.euro = Mediaan_bruto_inkomen_1000.euro * 4.5)
write.csv(af_newvariable, "data 2.0/Final_dataset_2.csv", row.names = FALSE)
```

Another variable that is valuable, especially in a comparison with the nominal maximum mortgage loan, is the real maximum mortgage loan. In order to calculate this, we set the year 2024 as the base year. In this year, the nominal value is equal to the real value. The real maximum mortgage loan is adjusted for inflation and clearly proves the money illusion and the sudden shocks. In order to create the extra column, it is necessary to use the cumprod() function, as the inflation has to be calculate cumulative.

```r
#af_extravariable <- af_newvariable %>%
#  arrange (Jaartal) %>%
#  mutate (prijs_index_2011 = cumprod((1 + Inflatie)/100 , )
```

3.3 Visualize temporal variation

3.4 Visualize spatial variation

Here you provide a description of why the plot above is relevant to your specific social problem.

3.5 Visualize sub-population variation

Here you provide a description of why the plot above is relevant to your specific social problem.

3.6 Event analysis

Analyze the relationship between two variables.

Here you provide a description of why the plot above is relevant to your specific social problem.

Part 4 - Discussion

4.1 Discuss your findings

Part 5 - Reproducibility

5.1 Github repository link

Provide the link to your PUBLIC repository here: ...

5.2 Reference list

https://www.dnb.nl/actuele-economische-vraagstukken/woningmarkt/onze-hoge-hypotheekschulden-risicos-en-oplossingen/
https://www.homefinance.nl/hypotheek/informatie/lti-hypotheek/

Use APA referencing throughout your document.